Experiment - 11

Aim:

To design and configure a simple internetwork using a router in Cisco Packet tracer

Procedure:

1. Open cisco Packet Tracer and create a new topology
2. Add 2 routers, switches and 3-4 PCs for each network.
3. Connect PCs to switches to routers using ethernet cables
4. Assign static IP address and subnet mask to all PCs
5. Configure IP address on router interface.

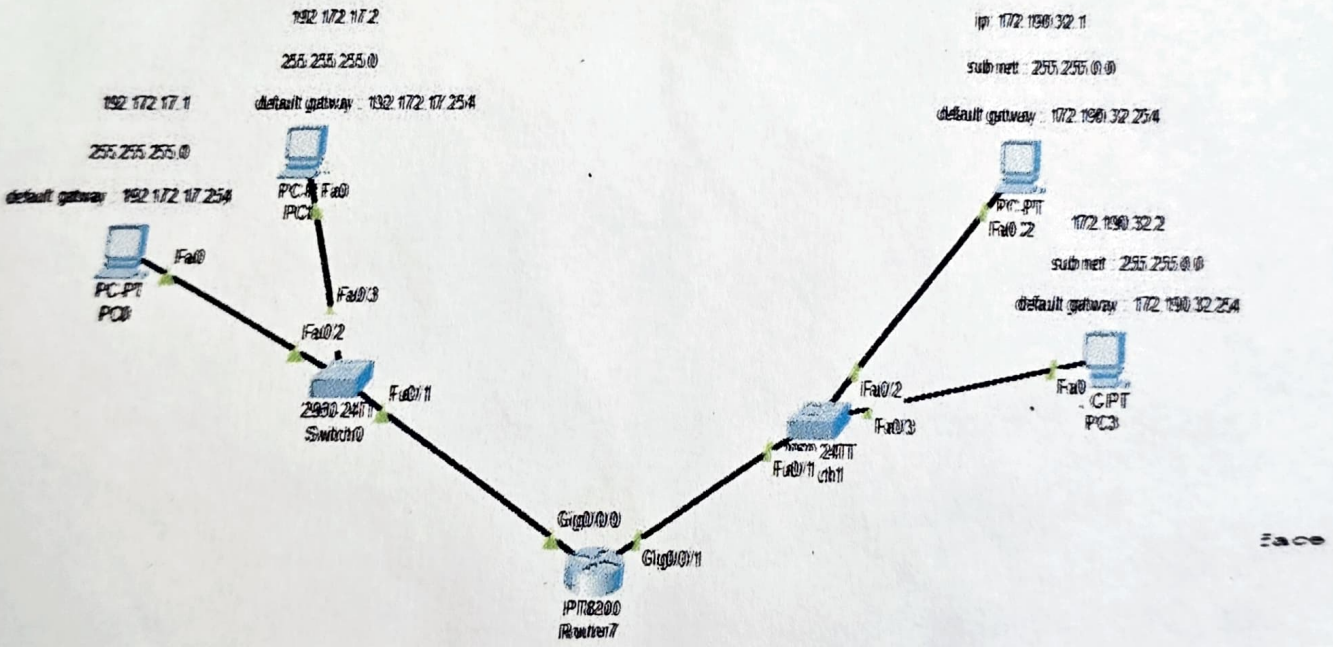
Code:

```
enable
configure terminal
interface FastEthernet 0/0
ip address <IP> <subnet mask>
no shutdown
exit
interface FastEthernet 0/1
IP address <IP> <subnet mask>
no shutdown
exit.
```

6. Configure default gateway of each PC with IP address of connected router interface

7. Use simulation mode and send a PDU from one network to another

8. Observe the packet path through the router and verify the successful communication



```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.172.17.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 172.190.32.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
```

8/10

Result :

The simple subnetwork was successfully designed and configured in cisco packet tracer and communication between different networks.